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Olist E-Commerce Fulfillment *Analysis*

Does delivery timing relative to estimated dates
drive measurable differences in customer review scores?

Dataset Olist Brazilian E-Commerce · Kaggle CC0
Scope 09/2016 – 08/2018 · 96,470 delivered orders
Tools SQL · Python · pandas · scipy · matplotlib
Date August 2026

Executive Summary

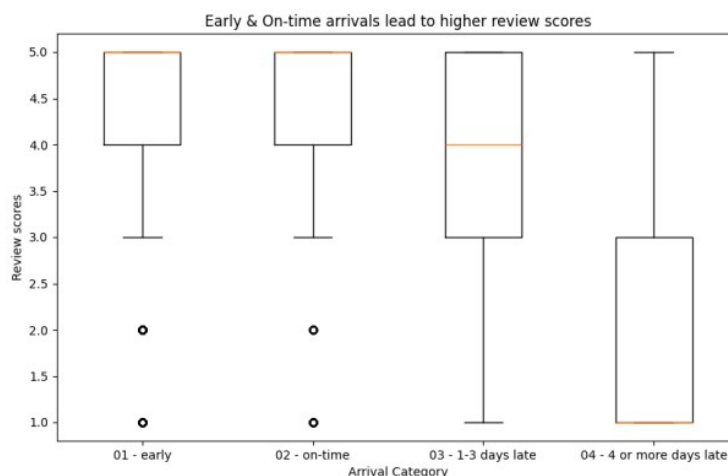
Testing reveals a very strong association ($p\text{-value} < 0.001$) between delivery timing relative to the estimated delivery date and review scoring.

Dataset & Scope

Data derived from the Olist e-commerce Public Dataset, published on Kaggle. The dataset ranges from 09/2016–08/2018. The scope of this analysis only covers orders with an order status of "delivered". After filtering on delivered, there were approximately 3,000 entries that had postal code mismatches that were omitted. This represents about 2% of the remaining 96,470 entries, that were marked "delivered".

Analytical Findings

The platform is doing well with on-time deliveries at 92.08%. Focusing on the opportunity area of late deliveries, when a delivery arrives 4 days or more later than expected, the customer review score plummets, on average to a 1.93 rating. By contrast, the early deliveries average a 4.3 rating.



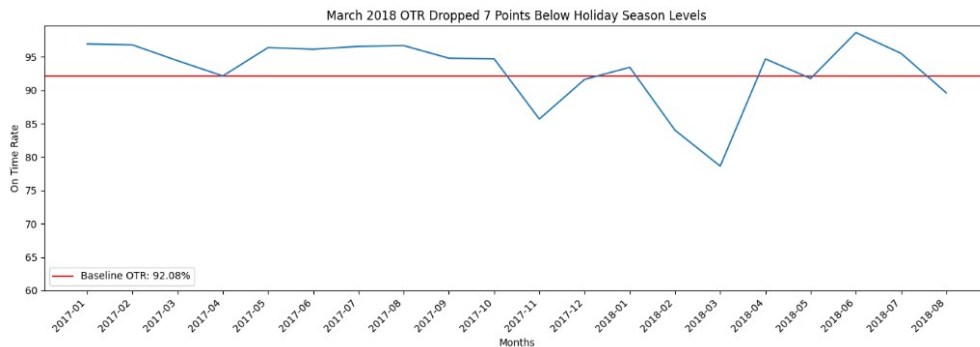
In an effort to determine what variables show a strong connection to late arrivals, a Pearson correlation test was done between the seller-customer distance and the corresponding review scores. The score came back at -0.077 , revealing there was effectively no correlation between these two variables.

Although it is clear that distance and delay connection is non-existent, we cannot overlook the fact that each state has its own OTR ranking. Following that thread, the state of MA did reveal itself as a bottom performer in the OTR category, by state, at 77%, this is notably 15 points below the platform average. With the understanding that distance is not a contributing factor in delays, the next question focused on shipping cost as a percentage of the weight.

Tests on shipping cost relative to freight weight was done to determine any correlations there. Again, testing revealed there was no meaningful correlation. Each seller was ranked based upon a

weighted composite score that segmented them based upon volume, on-time rate and review score. The composite was designed to balance these three measures and reveal a clear value ranking of each seller. Sellers were ranked into low, mid and high tiers. The composite was crafted by indexing the three measures & weighing them as: volume (0.45), OTR (0.40) & review score (0.15). Having analyzed down to the seller level, the analysis then turned to the delivery time variance over time.

Analysis shows holiday peak season OTR dips to 88.47%. This marks an almost 4 point drop in this two month peak season. The analysis reveals a more concerning anomaly, March 2018, with its 78.6% OTR. Making for a 13 point drop from platform average and almost 10 points worse than peak holiday season. The data is silent on the reasoning for the collapse, more study and a wider scope are needed to discern. Data collection ended in Aug 2018, the partial month data has been excluded from the dataset, so as not to misrepresent the month. With the analysis done on the seasonal OTR, the study looked to determine if there were any connections between the expected and actual arrival times and review scores.



Kruskal-Wallis test revealed a p-value < .001 and a test statistic of 10,089. These results confirm the association of arrival times and the review scores. Arrival times were bucketed into 4 ranges (early, on-time, 1-3 days late, 4+ days late). A Mann-Whitney U test similarly revealed a p-value < .001. This further confirmed the association between the expected arrival time (early vs. not early) and the resulting review scores.

Limitations

E-commerce sales data is limited to Brazil, only, during the period between 09/2016-08/2018. These findings are only representative of this scoping. No conclusions are to be drawn about other regions or other time frames.

The March 2018 anomaly of the 15 point collapse is only revealed through the data. There is no supporting evidence to reveal the source of the collapse.

With respect to the delivery delays, the data does not reveal any reasoning. There is no way to determine if the delivery left the seller late, got lost, or any other possibility. The only definite is the product arrived after the expected date.

Although the analysis reveals a strong association between delivery timing and review scores, the data does not establish a causal relationship between the two.

Appendix

Figure A — Seller-Customer Distance vs. Delivery Time Difference (Pearson coef: -0.077)

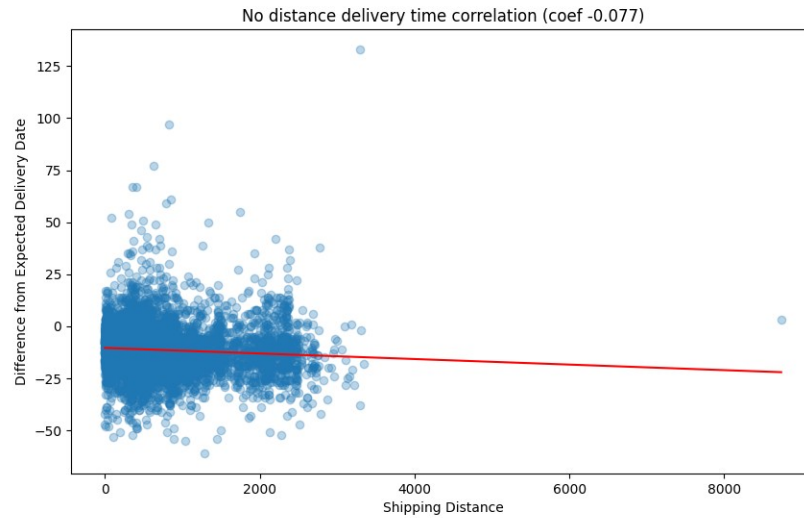


Figure B — Early vs. Not Early Arrival Review Score Distributions (Mann-Whitney U, $p < .001$)

